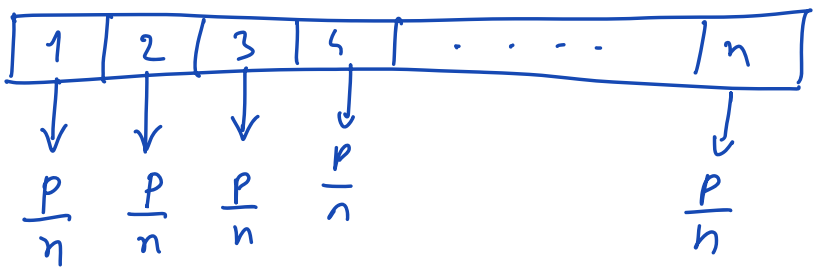
Avg. case of the Sequential Search

Assumptions: 1. The probability of "K is in A" is p.

2. The probability of K existing in any cell of A is the same.

3. K exists in only single location.



$$\frac{p}{n}\cdot 1 + \frac{p}{n}\cdot 2 + \frac{p}{n}\cdot 3 + \ldots\ldots + \frac{p}{n}\cdot n + (1-p)\cdot n$$

$$= \frac{p}{n}\cdot\frac{n(n+1)}{2} + n - np = \frac{pn}{2} + \frac{p}{2} + n - np = n + \frac{p}{2} - \frac{np}{2}$$

$$= n - p\left(\frac{n-1}{2}\right)$$

Order of growth:

$\log_2 n$ vs $\log_3 n$

$$\lim_{n\to\infty} \frac{\log_2 n}{\log_3 n} = \frac{\frac{\log n}{\log 2}}{\lim_{n\to\infty}\frac{\log n}{\log 3}} = \log_2 3 \Rightarrow c$$

$\underbrace{100n^5 + 35n^4 + 70n^3 + 100n^2 + 10^3}_{f(n)}$ vs $\underbrace{n^5 + 1}_{g(n)}$

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 100 \neq 0 \Rightarrow$ constant

They the same asymptotic order of growth.

2^n vs 3^n

$\lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0$ 3^n has higher as. growth.

$n!$ vs $(n-1)!$

$\lim_{n\to\infty} \frac{n!}{(n-1)!} = \lim_{n\to\infty} \frac{(n-1)!\, n}{(n-1)!} = \infty$

$n!$ is faster.

n^n vs n^{n+1}

$\lim_{n\to\infty} \frac{n^n}{n^{n+1}} = \lim_{n\to\infty} \frac{n^n}{n \cdot n^n} = 0$ n^{n+1} is faster

$5^{\frac{n}{2}}$ vs $2^n \cdot n$

$$\lim_{n\to\infty} \frac{5^{\frac{n}{2}}}{n \cdot 2^n} = \lim_{n\to\infty} \frac{\left(\sqrt{5}\right)^n}{n \cdot 2^n} = \lim_{n\to\infty} \left(\frac{\sqrt{5}}{2}\right)^n \cdot \frac{1}{n} = 0$$

$5^{n/2}$ is faster.

Example 2: Element Uniqueness Problem.

$$C_w(n) = \sum_{i=0}^{n-2} \underbrace{\sum_{j=i+1}^{n-1} 1}_{} = \sum_{i=0}^{n-2} \left(n - 1 - (i+1) + 1\right)$$

$$= \sum_{i=0}^{n-2} (n-1-i) = (n-1) + (n-2) + (n-3) + \cdots + 1$$

$$= \frac{n(n-1)}{2} \in \Theta(n^2)$$

Example 4 : Gaussian Elimination

$$T(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} \underbrace{\sum_{k=i}^{n} 2}_{2(n-i+1)} = 2 \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} (n+1-i)$$

$$= 2 \sum_{i=0}^{n-2} (n-1-(i+1)+1)(n+1-i)$$

$$= 2 \sum_{i=0}^{n-2} (n-i-1)(n-i+1)$$

$$= 2 \sum_{i=0}^{n-2} \left[(n-i)^2 - 1 \right] = 2 \sum_{i=0}^{n-2} (n-i)^2 - 2 \sum_{i=0}^{n-2} 1$$

$$= 2 \underbrace{\left[n^2 + (n-1)^2 + (n-2)^2 + \ldots + 2^2 \right]}_{} - 2(n-1)$$

$$= 2 \left[\frac{n(n+1)(2n+1)}{6} - 1 \right] - 2(n-1) \approx \Theta(n^3).$$

Example 1 Recursive evaluation of $n!$

$M(0) = 0 \qquad M(n) = M(n-1) + 1$

Let's apply backward substitution:

$$M(n) = M(n-1) + 1$$
$$= (M(n-2)+1) + 1 = M(n-2) + 2$$
$$= (M(n-3)+1) + 2 = M(n-3) + 3$$
$$\vdots$$
$$= M(n-n) + n = M(0) + n = 0 + n = n \qquad \in \Theta(n)$$

iterative alternative:

```
If n = 0 return 1
fact ← 1
for i := 2 to n
    fact ← fact * i
return fact.
```

$M(n) = \sum_{i=2}^{n} 1 = n-1 \in \Theta(n)$

aside:

if iterative & recursive alg. have the same asymptotic, better to use iterative.

Because in recursive, we have extra space asymptotic due to stack.

```
moveDisks(n) {
    moveDisks(n-1)    1st bar → 2nd bar
    moveDisks(1)      1st bar → 3rd
    moveDisks(n-1)    2nd → 3rd
}
```

$M(n) = \#$ moves

$M(1) = 1$

$M(n) = 2M(n-1) + 1$

Fibonacci:

$$r^2 - r - 1 = 0$$

$$r = \frac{1 \pm \sqrt{1 - 4 \cdot (-1)}}{2} \Rightarrow r_1 = \frac{1+\sqrt{5}}{2} \quad r_2 = \frac{1-\sqrt{5}}{2}$$

$$F(n) = \alpha \left(\frac{1+\sqrt{5}}{2}\right)^n + \beta \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$F(0) = 0 \quad F(1) = 1$

$F(0) = \alpha \cdot 1 + \beta \cdot 1 = 0 \qquad \alpha = -\beta \qquad \beta = -\alpha$

$F(1) = \alpha\left(\frac{1+\sqrt{5}}{2}\right) + \beta\left(\frac{1-\sqrt{5}}{2}\right) = 1$

$= \alpha\left(\frac{1+\sqrt{5}}{2}\right) - \alpha\left(\frac{1-\sqrt{5}}{2}\right)$

$= \cancel{\frac{\alpha}{2}} + \frac{\alpha\sqrt{5}}{2} - \cancel{\frac{\alpha}{2}} + \frac{\alpha\sqrt{5}}{2} = \alpha\sqrt{5} = 1$

$\alpha = \frac{1}{\sqrt{5}} \qquad \beta = \frac{-1}{\sqrt{5}}$

$$F(n) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n \;//$$

```
Fibb(int n) {
    if n=0  return 0
    else if n=1  return 1
    else  return  Fibb(n-1) + Fibb(n-2)
}
```

Basic op: addition

$A(0) = 0 \qquad A(1) = 0$

$A(n) = A(n-1) + A(n-2) + 1$

$A(n) - A(n-1) - A(n-2) = 1$ → not homogeneous.

$B(n) = A(n) + 1$ $\quad A(n) = B(n) - 1$

$B(n) - 1 - B(n-1) + 1 - B(n-2) + 1 = 1$

$B(n) - B(n-1) - B(n-2) = 0$

$B(0) = 1$

$B(1) = 1$

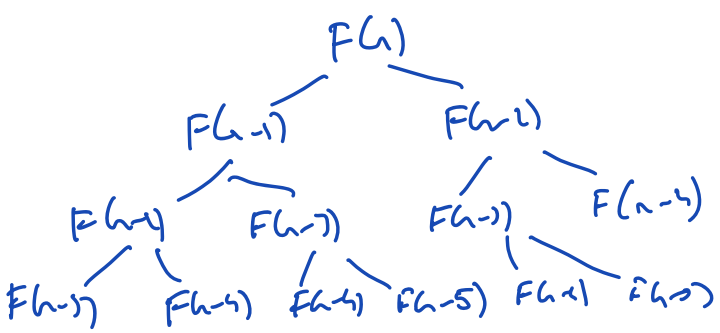
$B(n) = B(n-1) + B(n-2)$ → same as the fibonacci series, it starts from

$B(n) = F(n+1)$. $\quad B(0) = F(1)$. 1 1 2 3 5 ...

$$B(n) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^{n+1}$$

$$A(n) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^{n+1} - 1.$$

$A(n) \in \Theta(\phi^n)$ where $\phi = \frac{1+\sqrt{5}}{2}$ or $\phi \approx 1.6$



Iterative alg:

```
int fibb_iterative (int n)
    if n<0 return 0;
    else if n=1 return 1;
    F[0]←0;  F[1]←1;
    for i = 2 to n {
        F[n] ← F[n-1] + F[n-2]
    }
    return F[n].
```

$$A(n) = \sum_{i=2}^{n} 1 = n-1 \in \Theta(n).$$